MILITARY SPECIFICATION SHEET

SLEEVING, FOR IDENTIFICATION MARKING HEAT SHRINKABLE, POLYOLEFIN, FLEXIBLE

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the sleeving described herein shall consist of this document and the issue in effect of MIL-S-85848(AS).

REQUIREMENTS:

Continuous operating temperature range: Type I: -40°C to +105°C (-40°F to +221°F)
Type II: -55°C to +135°C (-55°F to +275°F)

Classification: The heat shrinkable identification sleeving shall be furnished in the following Types, and classes, as specified:

Types.

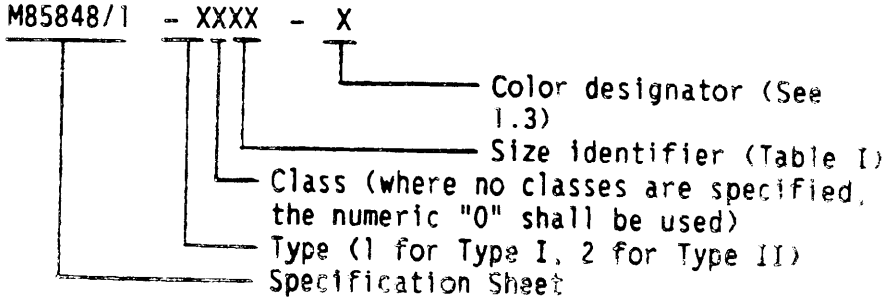
Type I - Flatweb format
Type II - Sleeving format

Classes.

Class 1 - Flame resistant
Class 2 - Flammable

Color: The standard color shall be white. Other colors are available. All colors shall conform to the requirements of Class 1 of MIL-STD-104.

Military Part Number: The Military Part Number shall consist of the basic number of this specification sheet and dash numbers as shown below:



AMSC N/A
FSC 5970

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Part number example: White, type I, class 1, .500 width, 1,000 length identification sleeving shall be identified as M85848/1-1104-9.

TABLE I - CONSTRUCTION DETAILS FOR TYPE I(FLATWEB) (Inches)

MILITARY PART NUMBER	W WIDTH AS SUPPLIED	L LENGTH AS SUPPLIED	RANGE OF WIRE DIAMETER	MIN. # OF SLEEVES/ PACKAGE	MIN. # OF SLEEVES FROM 1 FT LENGTH
Class 1					
MXXXXX/1					
-1101-*	.250 ± .03	1.000 ± .03	.035 - .075	5000	192
-1102-*	.333 ± .03	1.000 ± .03	.075 - .130	5000	144
-1103-*	.375 ± .03	1.000 ± .03	.130 - .160	5000	128
-1104-*	.500 ± .03	1.000 ± .03	.160 - .235	5000	96
-1105-*	.750 ± .03	1.000 ± .03	.235 - .375	3000	64
-1106-*	.250 ± .03	2.000 ± .03	.035 - .075	2500	96
-1107-*	.333 ± .03	2.000 ± .03	.075 - .130	2500	72
-1108-*	.375 ± .03	2.000 ± .03	.130 - .160	2500	64
-1109-*	.500 ± .03	2.000 ± .03	.160 - .235	2500	48
-1110-*	.750 ± .03	2.000 ± .03	.235 - .375	1500	32
Class 2					
MXXXXX/1					
-1201-*	.250 ± .03	1.000 ± .03	.035 - .075	5000	192
-1202-*	.333 ± .03	1.000 ± .03	.075 - .130	5000	144
-1203-*	.375 ± .03	1.000 ± .03	.130 - .160	5000	128
-1204-*	.500 ± .03	1.000 ± .03	.160 - .235	5000	96
-1205-*	.625 ± .03	1.000 ± .03	.235 - .290	5000	76
-1206-*	.667 ± .03	1.000 ± .03	.290 - .335	5000	72
-1207-*	.750 ± .03	1.000 ± .03	.335 - .390	3000	64
-1208-*	.833 ± .03	1.000 ± .03	.390 - .440	3000	56
-1209-*	.250 ± .03	2.000 ± .03	.035 - .075	2500	96
-1210-*	.333 ± .03	2.000 ± .03	.075 - .130	2500	72
-1211-*	.375 ± .03	2.000 ± .03	.130 - .160	2500	64
-1212-*	.500 ± .03	2.000 ± .03	.160 - .235	2500	48
-1213-*	.625 ± .03	2.000 ± .03	.235 - .290	2500	38

* The asterisk in the part number shall be replaced by color code designations.

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Table II Construction Details for Type II Sleeving

MILITARY PART NUMBER	DIAMETER AS SUPPLIED MIN	DIAMETER AFTER RECOVERY MAX.	LENGTH AS SUPPLIED	MINIMUM MARKING LENGTH	RANGE OF WIRE DIAMETER	NUMBER OF SLEEVES/UNIT PACKAGE, MIN
-2101-*	.093	.046	1.65±.05	1.5	.050-.085	5000
-2102-*	.125	.062	1.65±.05	1.5	.075-.115	5000
-2103-*	.125	.046	1.66±.08	1.5	.050-.115	5000
-2104-*	.187	.093	1.62±.05	1.5	.100-.165	5000
-2105-*	.187	.062	1.66±.08	1.5	.050-.165	2500
-2106-*	.250	.125	1.60±.05	1.5	.135-.235	2500
-2107-*	.250	.093	1.60±.08	1.5	.100-.235	2500
-2108-*	.375	.187	1.59±.08	1.5	.200-.355	1500
-2109-*	.375	.125	1.80±.08	1.5	.135-.355	1500
-2110-*	.475	.250	1.73±.13	1.5	.260-.450	1000
-2111-*	.093	.046	1.89±.05	1.75	.050-.085	5000
-2112-*	.125	.062	1.89±.05	1.75	.075-.115	5000
-2113-*	.125	.046	1.89±.05	1.75	.050-.115	5000
-2114-*	.187	.093	1.88±.05	1.75	.100-.165	2500
-2115-*	.187	.062	1.89±.08	1.75	.050-.165	2500
-2116-*	.250	.125	1.85±.05	1.75	.135-.235	2500
-2117-*	.250	.093	1.84±.05	1.75	.100-.235	2500
-2118-*	.375	.187	1.84±.05	1.75	.200-.355	1500

* The asterisk in the part number shall be replaced by color code designations

TABLE III - PHYSICAL PROPERTIES 1/

PROPERTY	REQUIREMENT	TEST PROCEDURE AND CONDITIONS
Dimensions	Table I	4.6.2
Fungus resistance	Rating of 1 or less	4.6.1 ASTM G21
Specific Gravity, max.	1.35	4.6.1 ASTM D792
Water Absorption,%,max.	1.0	4.6.1 ASTM D570 24 hrs @23°C(73°F)
Color stability	Pass	24Hrs @ 149°±1°C (300°±2°F)
Tensile strength, psi, min	Type I 1100 Type II-1500	4.6.5, ASTM D638 20 inch/min

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TABLE III - PHYSICAL PROPERTIES(Continued) 1/

PROPERTY	REQUIREMENT	TEST PROCEDURE AND CONDITIONS
Secant modulus, psi, max	TypeI-54,000 Type II-25,000	4.6.6.1, ASTM D882, 2% strain
Heat shock	No cracks, flowing or dripping 2/	4.6.7, Type I 121°±1°C(250°±F) Type II-250°±2°C(482°±6°F)
Heat resistance Print performance	No cracking 2/ Pass	4.6.8, Type I 105°±1°C(221°±2°F) Type II 136°±2°C(277°±4°F for 168 hours
Low temperature flexibility	No cracking 2/	4.6.9, Type I -40°±1°C(-40°±2°F) Type II -55°±1°C(-67°±2°F
Fluid resistance Tensile strength, after psi, min Print performance	 1000 Pass	4.6.10
Flammability	Class 1 self ext. within 1 minute, no more than 25% of indicator flag burned or charred	4.6.11.1
Print performance	Pass	4.6.13

1/ Unless otherwise specified, the stated requirement, test conditions and procedures are for all types and classes.
2/ The degree of bend shall be 90° up to 0.625 inch sleeve width or diameter. All others shall be 45°.

Restricted Shrinkage: Test method 4.6.3, Type I: 149°±2°C (300°±4°F) for 3 to 5 minutes until the sleeve is snug on the wire; Type II: 175°±1°C (347°±2°) for 5 minutes.

Unrestricted shrinkage: 4.5.1, Type I: 149°±2°C (300°±4°F) for 5 minutes; Type II: 200°±2°C (392°±5°F) for 3 minutes.

Shelf Life Conditions: Supplier shall certify to storage at ambient storage conditions for two years followed by conformance to the Construction Details of either Table I or Table II.

Shelf life extension: 4.6.13

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Intended use. Heat shrinkable polyolefin identification sleeving is used as a snug fitting non-insulating material to identify wires, wire bundles, cables and cable harnesses.

NOTICE

1. Flammability - Class 2 heat shrinkable identification sleeving described in this specification sheet has not been flame retarded and will burn readily. Class 2 sleeving shall not be considered for procurement when flame resistance is required.

Preparing Activity:
Navy AS
(Project No. 5970-N664-1)

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